

Understanding the GGOV1 Gas-Turbine Governor

Six controller tests and a novice-oriented fleet study

DataCenter repository

July 27, 2026

Abstract

This report explains the six educational GGOV1 simulations generated by `tools/vv/plot_ggov1_explainer.py`. It is written for a reader who has no previous experience with power-system frequency, synchronous generators, gas turbines, or turbine governors. The report first builds the physical picture—electrical load applies braking torque, fuel creates mechanical torque, and rotor speed determines electrical frequency—and then explains each test case in cause-and-effect order. The cases demonstrate a normal load increase, GGOV1’s low-value selection logic, droop versus isochronous control, thermal-limit takeover, acceleration protection after a load rejection, and comparison with published Hannett and Khan benchmarks. The figures are explanatory plots, not proof that the model is a calibrated LM2500 digital twin. The final section states what each case verifies, what it does not validate, and how to reproduce the results. A separate fleet-study chapter explains how `tools/vv/fleet_study.py` constructs a two-hour AI-workload trace, represents N equally sharing machines as one dynamic equivalent, screens $N = 2, 3$, and 4 for frequency ride-through, and estimates per-shaft torsional fatigue only for physically valid trajectories.

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1 The physical system in plain language

1.1 A gas turbine, generator, and electrical load

A gas turbine burns fuel and produces torque on a rotating shaft. A synchronous generator converts that shaft torque into electrical power. The electrical load—for example, a data center—acts like a braking torque on the same shaft. Stable operation requires a balance:

$$\text{mechanical power from turbine} \approx \text{electrical power consumed by load.} \quad (1)$$

The symbols used throughout this report are

$$P_m : \text{mechanical turbine power,} \quad (2)$$

$$P_e : \text{electrical power,} \quad (3)$$

$$\omega : \text{rotor speed in per unit,} \quad (4)$$

$$f = 60\omega : \text{electrical frequency in hertz for this model.} \quad (5)$$

At nominal speed, $\omega = 1$ pu and $f = 60$ Hz.

Plain-language explanation: Why frequency moves

When the load suddenly asks for more electrical power, the turbine cannot increase fuel and torque instantly. For a short time $P_e > P_m$. The missing energy comes from the kinetic energy stored in the spinning rotor, so the rotor slows and frequency falls. When load suddenly decreases, $P_m > P_e$ for a short time, so the rotor accelerates and frequency rises. Frequency is therefore a visible symptom of the mechanical–electrical power mismatch.

1.2 The swing equation

The simplified rotor dynamics in the repository are

$$2H \frac{d\omega}{dt} = P_m^{(g)} - P_e^{(g)} - D(\omega - 1), \quad (6)$$

where H is the inertia constant in seconds, D is machine damping, and the superscript (g) means generator-base per unit. Equation (6) says:

- if $P_m^{(g)} = P_e^{(g)}$, speed is not being accelerated by a power imbalance;
- if $P_m^{(g)} < P_e^{(g)}$, $d\omega/dt < 0$ and frequency falls;
- if $P_m^{(g)} > P_e^{(g)}$, $d\omega/dt > 0$ and frequency rises;
- larger inertia H makes speed change more slowly for the same power imbalance.

1.3 What the governor does

The governor is an automatic feedback controller. It measures speed and other signals, decides how much fuel is safe and useful, and commands a fuel valve. The repository implementation contains three competing controllers:

1. **Speed/power governor (FSRN).** This is the normal operating controller. It asks for more fuel when speed or measured power is too low and less fuel when speed is too high.
2. **Acceleration limiter (FSRA).** This protective controller reduces fuel when positive rotor acceleration is excessive, especially after a large load rejection.

3. **Temperature/load limiter (FSRT).** This protective controller limits sustained fuel so the turbine does not remain above its thermal loading reference.

GGOV1 uses a low-value gate:

$$\text{FSR} = \min(\text{FSRN}, \text{FSRA}, \text{FSRT}). \quad (7)$$

The lowest request wins because every protective controller must be able to reduce fuel even if the normal speed controller wants more.

Plain-language explanation: Why select the lowest request?

Imagine three people controlling a faucet. The normal operator requests the water needed to meet demand. Two safety inspectors specify maximum safe openings. The actual opening must obey the smallest request; otherwise one of the safety limits would be violated. GGOV1 treats fuel similarly.

1.4 Valve and turbine dynamics

The selected request FSR is not the physical valve position. The actuator has a lag, opening and closing rate limits, and minimum/maximum stroke limits. In simplified notation,

$$\dot{v} = \text{clip}\left(\frac{\text{clip}(\text{FSR}, V_{\min}, V_{\max}) - v}{T_{act}}, R_{close}, R_{open}\right), \quad (8)$$

where v is valve stroke. With the default LM2500 override, fuel flow is speed-dependent:

$$W_f = v\omega. \quad (9)$$

After the turbine lead-lag dynamics, mechanical power is approximately

$$P_{m,t} = K_{turb}(x_{turb} - W_{fnl}), \quad (10)$$

where W_{fnl} is the full-speed, no-load fuel requirement.

1.5 Per-unit bases used by the implementation

A per-unit value is a physical quantity divided by a chosen base. This model uses two bases:

- turbine/governor base: $T_{rate} = 22 \text{ MW}$;
- generator/swing-equation base: $S_n = 23 \text{ MVA}$.

The conversion is

$$k_b = \frac{T_{rate}}{S_n}, \quad P^{(g)} = k_b P^{(t)}. \quad (11)$$

Valve stroke, fuel flow, controller requests, and turbine mechanical power use the turbine base. Electrical power in the swing equation uses the generator base. Mixing these bases without Equation (11) would produce a numerically plausible but physically inconsistent result.

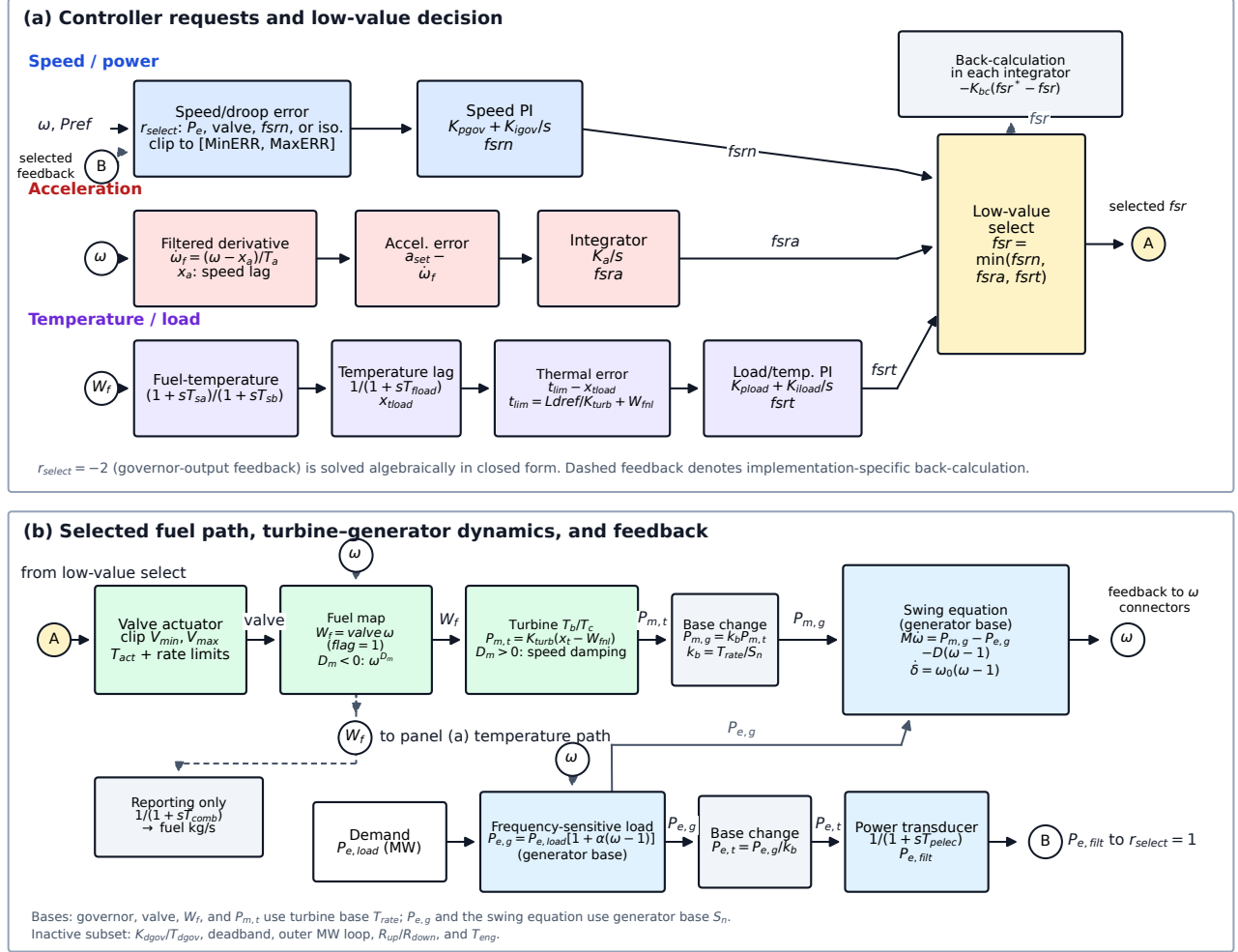


Figure 1: Implementation-oriented GGOV1 block diagram. Panel (a) shows the three controller requests and low-value decision. Panel (b) shows the selected valve/fuel/turbine path, generator dynamics, electrical-load path, and named feedback connectors. Solid arrows are forward decision flow; dashed arrows are feedback or reporting paths.

2 How the six test cases were constructed

All scenarios are generated by `tools/vv/plot_ggov1_explainer.py` using the implementation in `gas_plant/dynamics/ggov1.py`. Unless a case says otherwise, the parameters come from `GGOV1Params.lm2500_overrides()`: IEEE PES-TR1 Appendix-C typical values with a faster $T_{act} = 0.15\text{s}$ actuator and $R = 0.04$ droop. Table 1 summarizes the disturbances.

Table 1: Summary of the six explanatory cases.

Case	Scenario	Main behavior	Principal question
1	Load step 11.5 MW $\rightarrow$ 14 MW	Power mismatch, frequency nadir, recovery	Why does frequency fall after load increases?
2	Same 2.5 MW load increase	FSRN/FSRA/FSRTarbitration and valve lag	Which controller actually determines fuel?
3	Same load step in droop and isochronous modes	Different steady-state frequency objectives	Why does droop retain a frequency offset?
4	Overload 15 MW $\rightarrow$ 25 MW	Transient headroom followed by thermal-limit takeover	Why is short-term power not sustained capability?
5	Load rejection 15 MW $\rightarrow$ 8 MW	Overspeed tendency and acceleration protection	How does GGOV1 cut fuel after load disappears?
6	Hannett Table-3-style benchmark	Response-speed and speed-excursion metrics	Is the modeled response in the neighborhood of published gas-turbine results?

Important limitation: Educational tests versus complete validation

Cases 1–5 are controlled numerical demonstrations. They verify that the implemented loops interact in a physically understandable way, but they are not measurements from an LM2500 and do not establish model-parameter accuracy. Case 6 supplies literature context, but even that comparison is limited by unpublished plant parameters and differences between machines. These figures should support explanation and software verification, not a claim of OEM-grade LM2500 certification.

3 Case 1: normal load increase and frequency recovery

3.1 Question and setup

At $t = 0$ on the figure’s relative time axis, electrical demand changes from 11.5 MW to 14 MW. The physical step is applied at $t = 5\text{s}$ in the simulation; subtracting five seconds simply makes the plot easier to read. The plotted window emphasizes the first ten seconds, where the important governor dynamics occur.

3.2 Expected sequence

1. Electrical demand changes almost instantaneously.
2. Mechanical turbine power initially remains near its old value because the controller, actuator, fuel flow, and turbine all require time.
3. The negative power imbalance causes frequency to fall according to Equation (6).

4. The speed/power governor raises its fuel request. The valve opens at a finite rate and turbine mechanical power rises.
5. As P_m approaches P_e , the frequency decline stops at the *nadir*; frequency then recovers toward its controlled value.

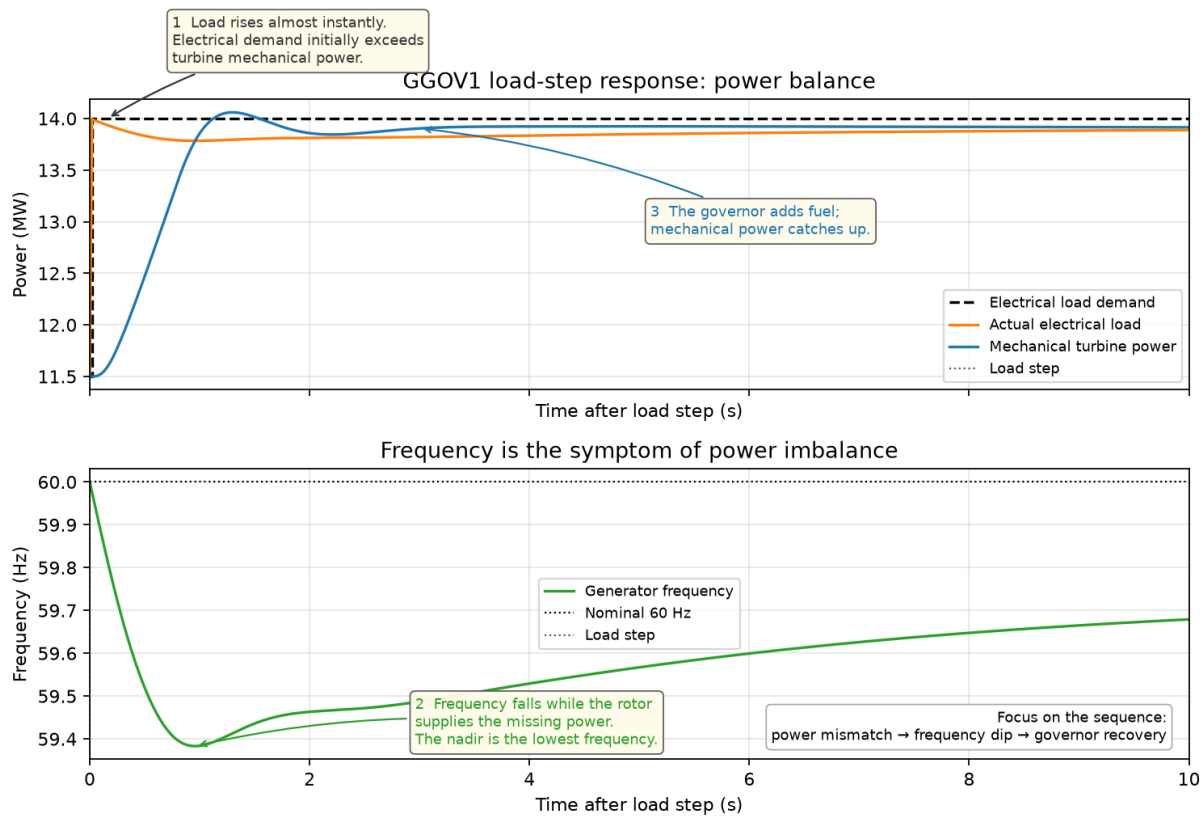


Figure 2: Case 1: cause-and-effect story for a normal load increase. The upper panel shows demand, actual frequency-sensitive electrical load, and turbine mechanical power. The lower panel shows the resulting generator frequency. Numbered annotations give the intended reading order.

What to focus on: Case 1

Do not read the frequency dip as a separate phenomenon. Compare the upper and lower panels at the same time. Frequency falls precisely while mechanical power is below electrical power. The nadir occurs near the time when that imbalance is being removed. Also distinguish the dashed demand trace from the orange actual electrical-load trace: the latter includes the model's frequency-sensitive load damping.

3.3 What this case tests

This case checks the sign and ordering of the coupled dynamics: increased load must produce a frequency decline, governor fuel must increase, mechanical power must follow with a delay, and the frequency response must stabilize. A wrong sign in the swing equation or governor feedback would make frequency move in the wrong direction or diverge.

4 Case 2: low-value selection and valve lag

4.1 Why three request signals are shown

Each GGOV1 branch continuously computes the fuel stroke it would choose:

- FSRN: normal speed/power request, plotted blue;
- FSRA: acceleration-protection request, plotted red;
- FSRT: temperature/load-protection request, plotted purple.

The thick black line is the selected minimum from Equation (7). The green line is the physical valve stroke after Equation (8). These are not interchangeable: FSR is a command, while v is a state of a physical actuator.

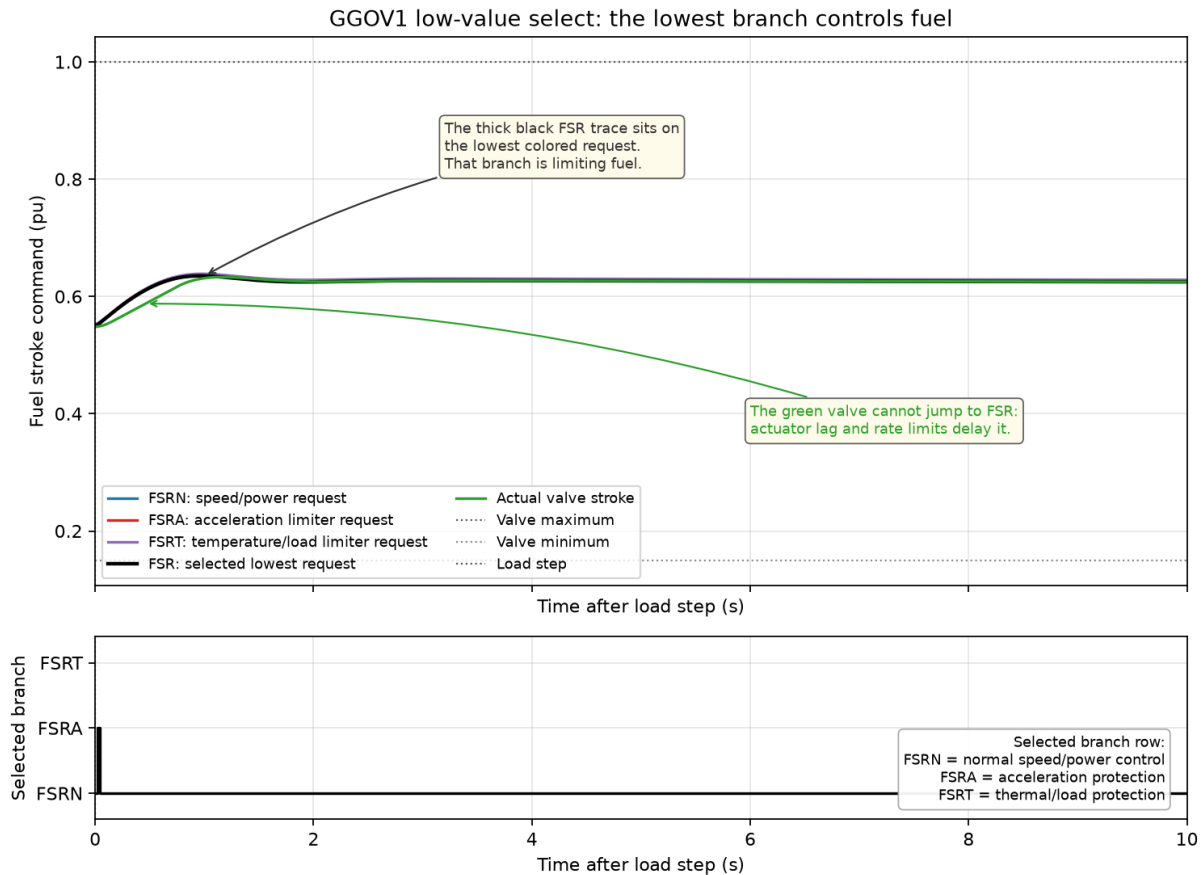


Figure 3: Case 2: GGOV1 controller arbitration during the same normal load increase as Case 1. The bottom panel converts the minimum calculation into a categorical view of the active branch.

What to focus on: Case 2

First find the thick black FSR line and identify which colored request lies under it. Then look at the categorical bottom panel for confirmation. Finally compare black FSR with green valve stroke. Their separation is the visible consequence of actuator lag and valve-rate limits; it is not a controller selection error.

4.2 Anti-windup behavior

When one branch is selected, the other controller integrators could otherwise continue accumulating error, or “wind up.” The implementation applies back-calculation:

$$\dot{x}_{igov} = K_{igov}e_{speed} - K_{bc,speed}(FSRN - FSR), \quad (12)$$

$$\dot{x}_a = K_a e_{accel} - K_{bc,accel}(FSRA - FSR), \quad (13)$$

$$\dot{x}_{iload} = K_{iload}e_{temp} - K_{bc,temp}(FSRT - FSR). \quad (14)$$

The tracking terms pull unselected branches toward the winning command so a later handoff is smoother.

4.3 What this case tests

The case checks that the numerical low-value gate agrees with the plotted active-branch category, that normal operation is not unintentionally dominated by a protective branch, and that valve motion respects its own dynamics rather than jumping directly to the controller request.

5 Case 3: droop versus isochronous control

5.1 Two different control objectives

The disturbance is again 11.5 MW to 14 MW, but the simulation is run twice.

Droop mode ($rselect = 1$). Electrical power passes through the T_{pelec} transducer and participates in the speed-governor error. In simplified form,

$$e_{speed} = (1 - \omega) + R(P_{ref} - P_{e,filt}). \quad (15)$$

Droop intentionally permits a steady frequency offset as load changes. This behavior allows multiple generators to share a load change without every controller trying to force the system independently to exactly 60 Hz.

Isochronous mode ($rselect = 0$). Power feedback is removed:

$$e_{speed} = 1 - \omega. \quad (16)$$

The integral controller continues adjusting fuel until steady frequency returns to nominal. Isochronous control is natural for one designated master unit in an island, but several independent isochronous governors can fight one another unless coordinated.

What to focus on: Case 3

Focus on the final values, not only the initial dip. The blue droop trace settles with a deliberate frequency error; the green isochronous trace returns toward 60 Hz. This does not make isochronous control universally “better.” Droop provides decentralized power sharing, while isochronous control provides exact frequency restoration for a suitable master unit.

5.2 What this case tests

This test exercises two code paths selected by `rselect`. It confirms that electrical-power droop feedback is active for $rselect = 1$ and omitted for $rselect = 0$, and that the long-term responses reflect those different control laws.

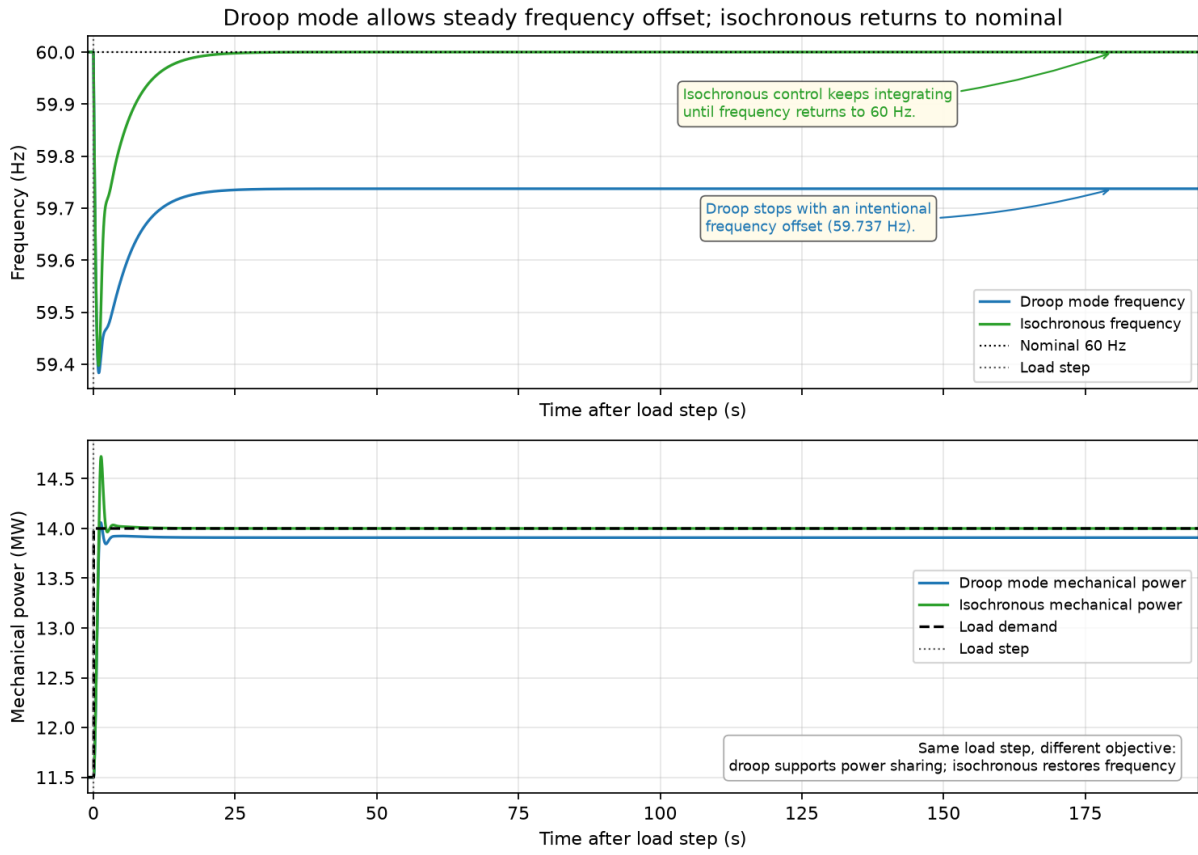


Figure 4: Case 3: frequency and mechanical-power response for droop and isochronous control. Both cases stabilize, but their steady-state objectives differ.

6 Case 4: overload request and thermal-limiter takeover

6.1 Setup and two different power ceilings

The requested load increases from 15 MW to 25 MW. This request is above the model's continuous thermal reference:

$$P_{thermal,max} = Ldref T_{rate} = 1.0 \times 22 \text{ MW} = 22 \text{ MW}. \quad (17)$$

The valve can temporarily produce a higher idealized mechanical ceiling:

$$P_{transient,max} = K_{turb}(V_{max} - W_{fnl})T_{rate} = 1.5(1.0 - 0.2) \times 22 \text{ MW} = 26.4 \text{ MW}. \quad (18)$$

The difference between 26.4 MW transient capability and 22 MW continuous capability is central to the case.

6.2 Temperature-path logic

The implementation uses fuel flow as a temperature/load proxy. It passes through a lead-lag and a lag, then compares with the fuel level corresponding to the thermal reference:

$$t_{lim} = \frac{Ldref}{K_{turb}} + W_{fnl}, \quad e_{temp} = t_{lim} - x_{tload}, \quad (19)$$

$$\text{FSRT} = K_{pload}e_{temp} + x_{iload}. \quad (20)$$

As the filtered fuel proxy rises, FSRT becomes more restrictive. Once it is below FSRN, the low-value gate selects it and sustained fuel is reduced.

What to focus on: Case 4

The 25 MW dashed load request is not a promise that the turbine can supply 25 MW indefinitely. Read the panels together: temporary valve headroom raises mechanical power, then purple FSRT falls below blue FSRN and becomes the selected request. Frequency remains low because protecting the turbine and satisfying an infeasible load request are conflicting objectives.

Important limitation: Temperature is represented by a proxy

This repository does not solve combustor metal temperature, exhaust-temperature thermocouple dynamics, blade cooling, or OEM thermal-control logic. The GGOV1 load-limiter path is a standard reduced-order control representation driven by fuel flow. The case verifies the implemented limiter architecture; it does not predict component temperature or prove an LM2500 overload duration.

7 Case 5: load rejection and acceleration protection

7.1 Setup

Electrical load falls from 15 MW to 8 MW. Immediately after the rejection, fuel and mechanical torque remain near their pre-event values, but electrical braking torque has decreased. Therefore $P_m > P_e$, the right side of Equation (6) is positive, and the rotor accelerates.

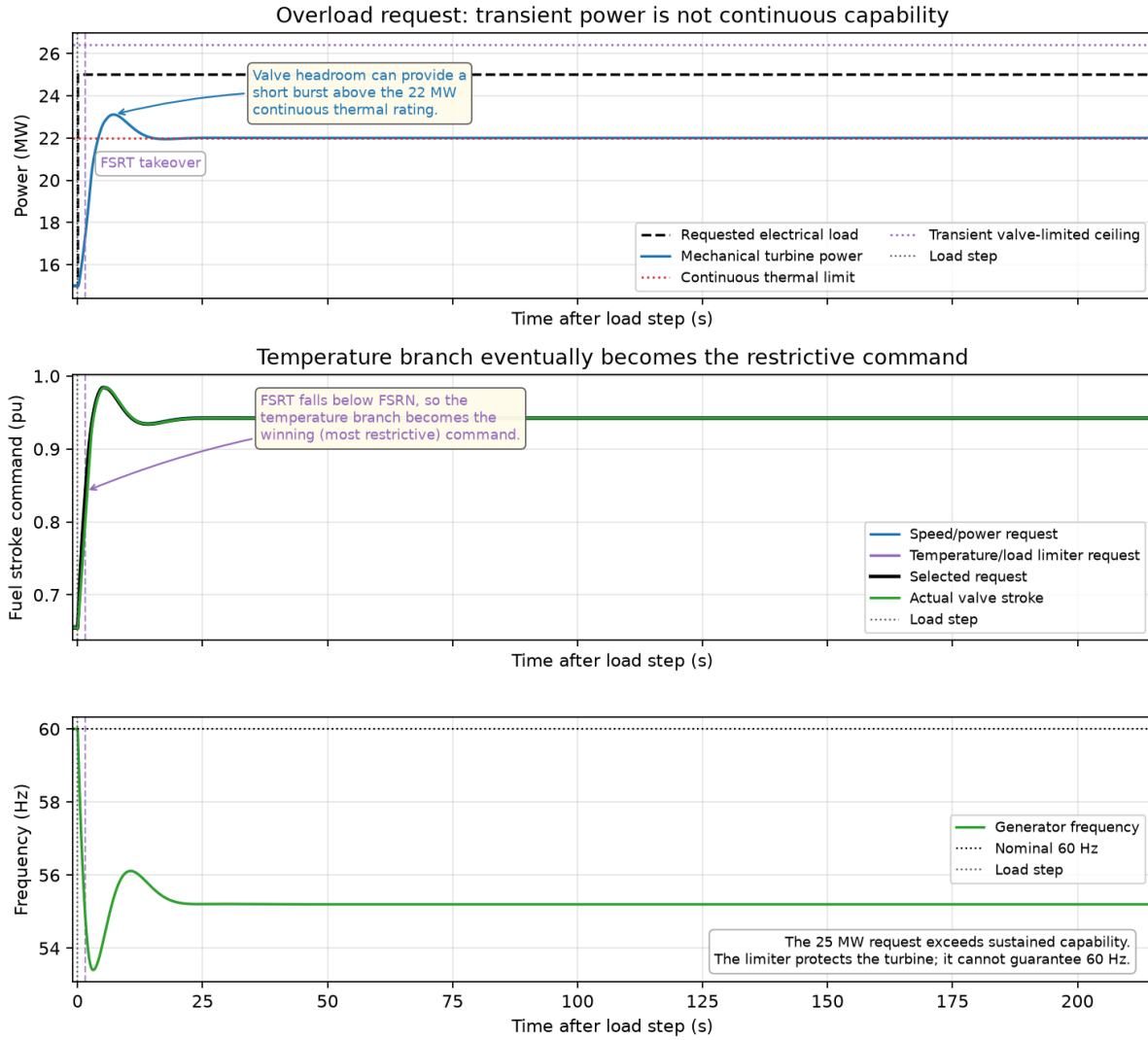


Figure 5: Case 4: an overload request produces transient mechanical-power headroom, followed by takeover of the temperature/load branch. The purple dashed vertical line identifies sustained FSRTselection.

7.2 Filtered acceleration and protective request

A raw numerical derivative of speed is noisy. GGOV1 uses a filtered derivative based on a first-order speed-lag state:

$$\dot{x}_{accel,lag} = \frac{\omega - x_{accel,lag}}{T_a}, \quad a_f = \frac{\omega - x_{accel,lag}}{T_a}. \quad (21)$$

The acceleration error is

$$e_{accel} = a_{set} - a_f, \quad (22)$$

and the acceleration branch is a pure integrator request:

$$\begin{aligned} \text{FSRA} &= x_a, \\ \dot{x}_a &= K_a e_{accel} \\ &\quad - K_{bc,accel}(\text{FSRA} - \text{FSR}). \end{aligned} \quad (23)$$

When acceleration exceeds its setpoint, $e_{accel} < 0$. This negative error drives FSRA downward.

If FSRA becomes the lowest request, the low-value gate commands a rapid fuel reduction.

What to focus on: Case 5

Follow the panels from the disturbance downward. Load rejection first creates excess mechanical torque. Positive acceleration appears before frequency has reached its maximum. During the shaded interval, red FSRA is the lowest request and black FSR follows it, cutting fuel. The green valve follows more slowly because physical actuator constraints still apply.

7.3 What this case tests

The case checks disturbance sign, filtered-derivative behavior, acceleration setpoint comparison, low-value selection of FSRA, and subsequent reduction of fuel. It is a structural protection test; it is not a certification of overspeed-trip settings or shaft-stress limits.

8 Case 6: published validation context

8.1 Why a literature comparison is needed

Cases 1–5 can show that software behaves coherently, but coherent behavior can still come from incorrectly tuned equations. Case 6 compares response metrics with Hannett and Khan’s published gas-turbine model-validation study. The protocol represented here starts at 50% of generator MVA rating and applies a 10% rating load increase, with 3% droop. For the repository’s $S_n = 23$ MVA generator, this corresponds to

$$P_0 = 0.5S_n = 11.5 \text{ MW}, \quad P_1 = 0.6S_n = 13.8 \text{ MW}. \quad (24)$$

8.2 The two benchmark metrics

Time to 0.6 pu mechanical power. This is the time after the disturbance at which mechanical power first reaches 0.6 pu on generator base. A shorter time indicates a faster turbine and governor response.

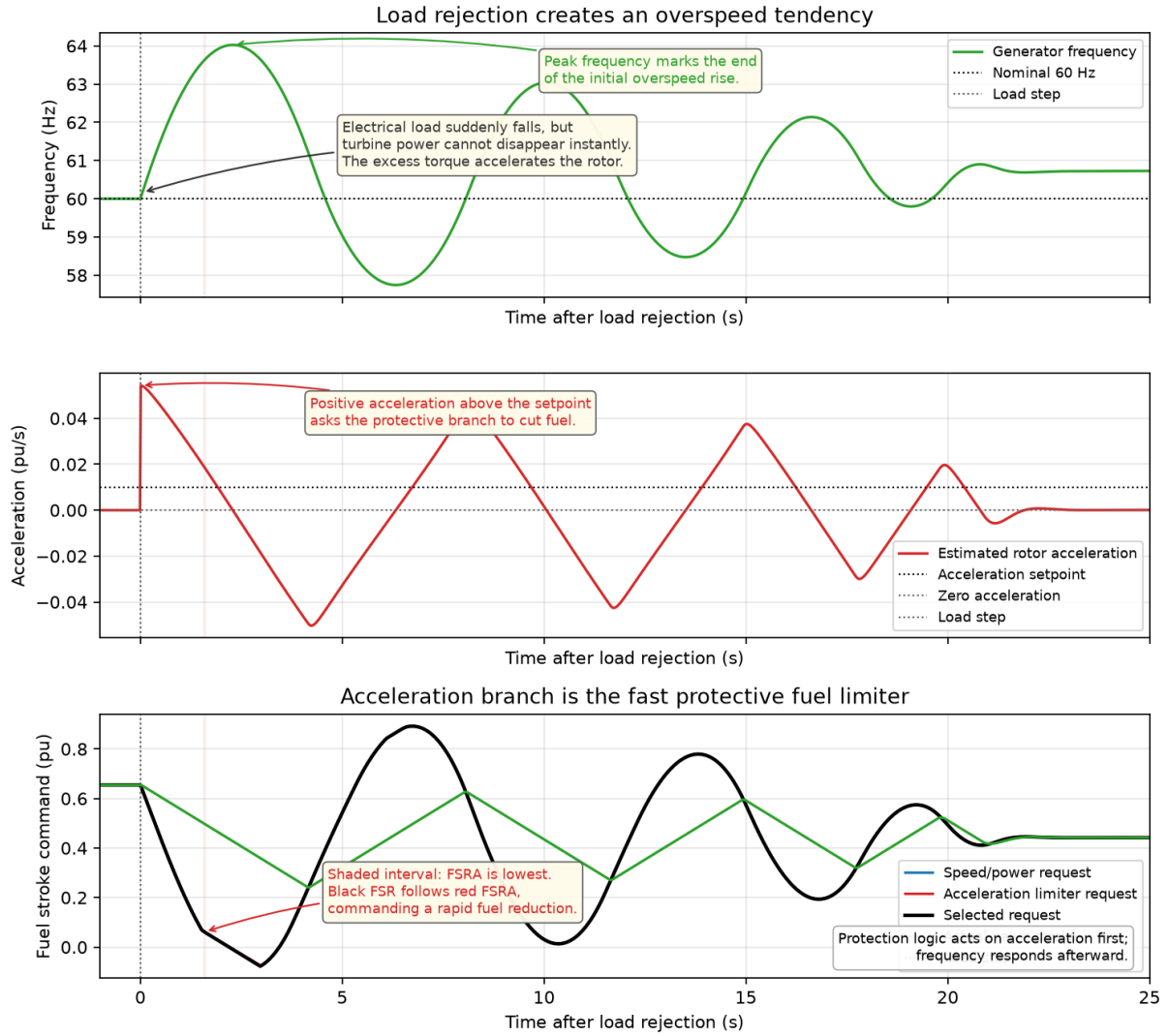


Figure 6: Case 5: frequency, estimated acceleration, and controller requests after a load rejection. Light red shading marks the interval during which the acceleration branch is selected.

Maximum rotor-speed excursion. For a load increase the excursion is negative because speed falls. A value closer to zero means a smaller frequency dip. At a 60 Hz base, for example, -0.004 pu corresponds to approximately $60(-0.004) = -0.24$ Hz.

The two published anchors plotted are

Published anchor	Time to 0.6 pu	Minimum speed excursion
Hannett typical model	1.140 s	-0.0039 pu
Hannett field-derived model	2.320 s	-0.0076 pu

The repository then runs GGOV1 with PES-TR1 typical values, GGOV1 with LM2500 overrides, and a Rowen typical-gas model under the same simplified protocol.

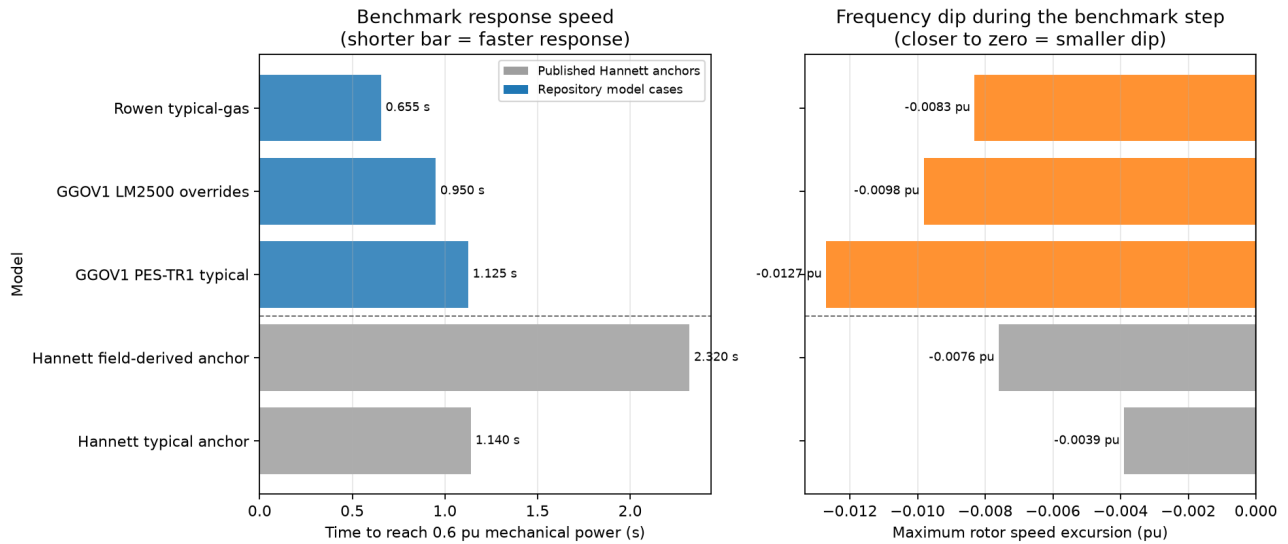


Figure 7: Case 6: published Hannett anchors (gray) and repository model cases (colored). The left panel compares response speed; the right compares the maximum speed/frequency dip. The dashed horizontal line separates published anchors from repository simulations.

What to focus on: Case 6

Do not select a model by matching only one bar. A model could reach mechanical power quickly by being unrealistically aggressive and simultaneously produce a wrong speed excursion. Credibility requires reasonable agreement in response time, speed excursion, waveform shape, and the parameter assumptions behind the comparison.

Important limitation: What the bars do not prove

The Hannett anchors are from other gas-turbine units and model parameter sets, not a controlled test on this repository's specific LM2500. Similar bars show that the response is in a plausible published neighborhood. They do not uniquely identify actuator constants, inertia, thermal limits, or an OEM controller implementation. Field data from the actual generator package would be required for high-confidence LM2500 calibration.

9 Fleet study: what changes when 2, 3, or 4 units share the load?

The six preceding cases explain one governor in controlled disturbances. The script `tools/vv/fleet_study.py` asks a different, practical question: how many identical LM2500-class units are needed to follow a rapidly changing data-center load, and what torsional fatigue does one shaft experience while doing so? This section explains the code from input data to reported results.

Plain-language explanation: What “fleet” means in this study

The script does not simulate several independently controlled generators and a detailed electrical network. It replaces N identical, equally sharing units on one common bus with one *aggregated equivalent machine*. Its power and MVA ratings are N times those of one unit, while its per-unit control parameters are unchanged. This is a standard screening approximation, but it cannot reveal unequal load sharing, circulating power, unit-to-unit oscillation, breaker behavior, or network voltage effects.

This dynamic aggregation is also separate from the steady-state `gas_plant/fleet.py` dispatch helper. The fleet study directly calls the GGOV1, multishaft, and torsional dynamic models.

9.1 Step 1: construct the two-hour electrical-load trace

The function `build_trace()` reads `data/nvidia_smi_first_1gb.parquet`. Each row is a measured GPU power sample. The code performs the following operations:

1. Sort all records by time and find the two-hour interval containing the densest measurement activity.
2. Create a 0.1s time grid. For each node and GPU, retain the last observation in a slot and hold that value until the next sample. This is a zero-order hold, not an interpolation that invents smooth ramps between measurements.
3. Sum the reconstructed GPU powers into a raw IT-load signal $B(t)$.
4. Apply a power-usage effectiveness value of 1.25. The added non-IT load is represented as 0.25 times the mean IT load.
5. Scale the resulting waveform so its maximum is 18 MW.

For this run, the constructed facility load spans 5.2 MW to 18.0 MW. Its largest downward and upward single-sample changes are -10.5 MW and $+9.6$ MW, respectively.

Important limitation: The trace is a conservative constructed scenario

The scaling treats the sampled GPU activity as if enough similar devices were perfectly synchronized to create an 18 MW peak. It is useful as a stress test, but it is not a measured 18 MW facility feeder trace. Real workload diversity, power-supply controls, UPS equipment, batteries, and network constraints may substantially change the waveform seen by a turbine.

9.2 Step 2: represent N turbines as one dynamic equivalent

The function `run_fleet()` creates a GGOV1 parameter set with

$$S_N = N(23 \text{ MVA}), \quad T_{rate} = N(22 \text{ MW}), \quad (25)$$

for $N = 2, 3$, and 4 . Thus the fleets have generator bases of 46 MVA, 69 MVA, and 92 MVA, and turbine ratings of 44 MW, 66 MW, and 88 MW. The turbine-to-generator base ratio remains

$$k_b = \frac{T_{rate}}{S_N} = \frac{22}{23}, \quad (26)$$

independent of fleet size.

The aggregated machine is passed to `simulate_multishaft()`. This Tier-C model has 13 dynamic states: the GGOV1 controller and turbine states, the power-turbine/generator rotor angle and speed, and an HP gas-generator speed. Its power-turbine/generator speed follows the familiar balance

$$2H_{PT} \frac{d\omega_{PT}}{dt} = P_{couple} k_b - P_e - D_{PT}(\omega_{PT} - 1). \quad (27)$$

The present model uses $H_{PT} = 2.5$ s and $D_{PT} = 0$; frequency-sensitive load is represented in P_e instead of being counted a second time as machine damping. Because the same MW disturbance is divided by a larger base as N grows, it becomes a smaller per-unit disturbance. The aggregate stored rotor energy and total valve-moving capability also increase with N . Therefore a larger fleet should exhibit a smaller frequency excursion.

The HP rotor is coupled to the power turbine by the reduced-order gas-path law

$$P_{couple} = K_{couple}(\omega_{HP} - \omega_{HP,idle}). \quad (28)$$

This is a linear placeholder, not a detailed compressor and turbine map.

9.3 Step 3: select a demanding 15-minute screening interval

The script locates the largest absolute one-sample load change and extracts 900 s around it. Each N value is first simulated on this same segment with a 0.02 s output interval. The `report()` function computes:

- minimum and maximum electrical frequency;
- total sampled time below 57.8 Hz, used here as an instantaneous-trip screening threshold; and
- time outside the 59.4 Hz to 60.6 Hz continuous operating band.

These are screening criteria rather than a complete relay model. In particular, “time below 57.8 Hz” is the sum of samples below the threshold; an actual protection system would normally trip at the first qualifying event and the later simulated trajectory would never occur.

9.4 Step 4: calculate per-shaft torsional fatigue only if the unit survives

For a surviving trajectory, `per_shaft_fatigue()` makes a shallow copy of the fleet result and divides mechanical and electrical power by N :

$$P_{m,unit} = \frac{P_{m,fleet}}{N}, \quad P_{e,unit} = \frac{P_{e,fleet}}{N}. \quad (29)$$

It then drives one 23 MVA two-mass shaft model. The power-turbine rotor and generator rotor are connected by a torsional spring and damper:

$$2H_{PT,o} \dot{\omega}_{PT,o} = \tau_{in} - \tau_{shaft}, \quad (30)$$

$$2H_{gen} \dot{\omega}_{gen} = \tau_{shaft} - \tau_e, \quad (31)$$

$$\tau_{shaft} = k\theta + c(\omega_{PT,o} - \omega_{gen}). \quad (32)$$

The assumed torsional mode is 22 Hz. The model samples shaft torque at 500 Hz, which is fast enough to resolve that mode.

A centered 5 s rolling median separates torque into a slow trend and a high-frequency residual. Rainflow counting converts each history into equivalent load cycles. Miner’s rule sums the fractional damage,

$$D = \sum_i \frac{n_i}{N_{f,i}}, \quad (33)$$

where n_i is the counted number of cycles and $N_{f,i}$ is the predicted number of cycles to failure at that amplitude. The Basquin relation used to obtain N_f has the form

$$N_f = N_{ref} \left(\frac{S_{a,ref}}{S_{a,eq}} \right)^m. \quad (34)$$

The high-cycle-fatigue (HCF) residual uses $m = 9$; the slow low-cycle-fatigue (LCF) trend uses $m = 4$. A Goodman correction increases the equivalent alternating stress when nonzero mean torque is present.

If every unit saw the same cycles and only the torque amplitude changed as $1/N$, Equation 34 would imply

$$D_{unit} \propto N^{-m}, \quad \frac{D_3}{D_4} \approx \left(\frac{4}{3} \right)^m. \quad (35)$$

The actual simulations need not obey this exactly because changing N also changes the governor and rotor trajectory, and therefore changes the cycles being counted.

9.5 Results for $N = 2$, $N = 3$, and $N = 4$

Table 2 reports the newly executed 15-minute screening cases. Fatigue is deliberately omitted for $N = 2$: it spends 235.1 s below the trip threshold, so the model’s later shaft history describes a generator that should already have disconnected.

Table 2: Worst-segment fleet screening results from `pixi run vv-fleet`.

N	Frequency range (Hz)	Below 57.8 Hz (s)	Out of band (s)	Verdict
2	40.526–66.600	235.1	527.2	Trip; fatigue skipped
3	59.249–60.830	0.0	14.6	Survives; band excursions
4	59.584–60.549	0.0	0.0	Passes segment screen

The progression is the central result. Doubling from two to four units does not merely double available MW. It also halves the same disturbance on a per-unit basis and doubles aggregate inertia on the enlarged base. The two-unit equivalent undergoes a severe frequency collapse and overshoot. The three-unit case avoids the trip threshold but leaves the continuous band for 14.6 s. The four-unit case remains within the selected continuous band throughout this segment.

Table 3 contains the valid per-shaft fatigue results. The “years to $D = 1$ ” column simply repeats the analyzed window continuously until accumulated Miner damage reaches one. It is an equivalent screening number, not a literal warranty life or predicted calendar failure date.

Table 3: Per-shaft fatigue metrics for segment trajectories that do not cross the trip threshold.

N	Torque range (kN m)	D_{HCF}	D_{LCF}	Equivalent years to $D = 1$
3	2.0–16.2	3.25×10^{-21}	1.27×10^{-11}	2.25×10^6
4	1.8–11.3	6.52×10^{-23}	3.45×10^{-12}	8.26×10^6

The measured LCF damage ratio is

$$\frac{D_{LCF,3}}{D_{LCF,4}} = 3.7, \quad \left(\frac{4}{3} \right)^4 = 3.2, \quad (36)$$

which is close to the simple amplitude-scaling prediction. The HCF ratio is 49.9, compared with $(4/3)^9 = 13.3$. This larger benefit does not mean that Basquin’s exponent changed. It means the four-unit dynamic trajectory also excites fewer or smaller high-frequency torque cycles than a pure rescaling of the three-unit waveform would predict.

What to focus on: How to interpret the fleet result

Read frequency compliance before reading fatigue. The $N = 2$ simulation crosses the trip threshold, so a tiny post-processed fatigue number would be misleading even if the code could calculate one. Among the surviving segment cases, calculated torsional damage is extremely small and $N = 4$ performs better than $N = 3$. Therefore this scenario is constrained first by frequency ride-through, not by the modeled PT-generator torsional fatigue.

9.6 What the full-window run does—and does not—establish

After screening the three fleet sizes, the current script runs only $N = 4$ on the complete two-hour trace. This saves substantial Tier-C and Tier-E simulation time while checking the segment’s best-performing unassisted candidate against events outside the selected 15-minute excerpt. The full-window $N = 4$ result spans 59.527 Hz to 60.675 Hz, never falls below 57.8 Hz, and spends 2.3 s outside the 59.4 Hz to 60.6 Hz band. Per-shaft torque spans 0.6 kNm to 11.7 kNm; $D_{HCF} = 5.60 \times 10^{-22}$ and $D_{LCF} = 2.76 \times 10^{-11}$, corresponding to 8.27×10^6 equivalent years to $D = 1$ under continuous repetition of this window.

A passing short segment cannot by itself prove that $N = 2$ or $N = 3$ would pass every other event; each candidate needed for a design decision should be tested on the full trace and on an ensemble of credible traces. Likewise, the small two-hour damage fraction is a model output under stated assumptions, not evidence for an eight-million-year physical service life.

9.7 Important modeling limits

The fleet results should be treated as comparative screening evidence because:

1. equal sharing is assumed; each machine is identical and sees the same speed, control tuning, and fraction of fleet power;
2. no buses, cables, transformers, excitation systems, reactive-power sharing, or generator relays are represented;
3. the HP-to-power-turbine gas-path law and 22 Hz torsional mode are placeholders pending machine-specific measurements;
4. absolute fatigue life depends strongly on assumed shaft geometry, steel properties, damping, rainflow separation, and S-N parameters;
5. the fatigue calculation covers PT-generator shaft torsion, not thermal fatigue from starts, stops, combustor cycling, or hot-section temperature changes; and
6. the extrapolated years assume continuous repetition of this one constructed window and therefore should not be used as a maintenance interval.

10 How the cases fit together

The six controller cases and the fleet study form a deliberate progression:

1. **Case 1 establishes the physical story:** imbalance moves frequency and governor action restores mechanical power.
2. **Case 2 opens the controller:** it shows which internal branch wins and why command and valve position differ.
3. **Case 3 changes the normal-control objective:** droop sharing and isochronous restoration are distinguished.
4. **Case 4 activates slow protection:** the thermal/load branch limits an unsustainable demand.
5. **Case 5 activates fast protection:** the acceleration branch cuts fuel during an overspeed tendency.
6. **Case 6 adds external context:** response metrics are compared with published gas-turbine anchors.
7. **The fleet study changes scale:** the same dynamic machinery is used to compare frequency ride-through and valid per-shaft fatigue for 2, 3, and 4 equally sharing units.

Table 4: What each case supports and what remains outside its scope.

Case	Evidence provided	Not established
1	Correct qualitative power/frequency causality and stable recovery	Exact LM2500 frequency nadir or field tuning
2	Correct low-value arbitration and actuator delay	OEM proprietary fuel-control details
3	Distinct droop and isochronous code paths	Multi-unit load sharing on a network
4	Thermal branch can cap sustained fuel	Metal temperature, emissions, or allowed overload duration
5	Acceleration branch can rapidly reduce fuel	Certified overspeed protection or trip-relay performance
6	Plausibility relative to published benchmark metrics	Unit-specific LM2500 validation
Fleet	Comparative effect of N on frequency and modeled shaft damage	Independent-unit sharing, detailed network behavior, or absolute service life

11 Model limitations that matter when reading the plots

1. **Reduced-order model.** GGOV1 is intended for power-system dynamic studies, not detailed combustion, aerodynamics, blade heat transfer, emissions, or compressor surge prediction.
2. **Single-machine electrical representation.** The explainer cases use a swing equation and frequency-sensitive aggregate load, not a multi-bus network with voltage, excitation, power flow, and relay dynamics.
3. **Typical and estimated parameters.** Most GGOV1 values are PES-TR1 typical values. The faster actuator and droop are LM2500 overrides, but the full parameter set has not been identified from a dedicated LM2500 transient test.

4. **Implemented subset.** The code does not implement several optional GGOV1 features: derivative action (K_{dgv}/T_{dgv}), deadband, the outer MW loop, valve-command limits (R_{up}/R_{down}), and engine transport delay (T_{eng}). These features are inactive in the typical parameter set used here.
5. **Protection versus trip systems.** GGOV1’s acceleration and load-limit branches are control limiters. They are not substitutes for independent overspeed, overtemperature, underfrequency, or generator-protection relays.

12 Reproducing the report figures

From the repository root, regenerate the annotated figures with

```
pixi run vv-ggov-plots
```

To also export the numerical time-series data used in Cases 1–5, run

```
pixi run python tools/vv/plot_ggov1_explainer.py --export-csv
```

The CSV files are written beneath `docs/figs/ggov1_explainer_signals/`. Regenerate the block diagram with

```
pixi run vv-ggov-block
```

Run the $N = 2$, 3, and 4 segment screen and the full-window $N = 4$ confirmation with

```
pixi run vv-fleet
```

The fleet task reads `data/nvidia_smi_first_1gb.parquet` and prints its frequency, operating-band, shaft-torque, and fatigue metrics to the terminal. It remains separate from `vv-all` because it is data-dependent and substantially longer than the unit tests. Compile this report from the repository root with

```
pixi run tectonic -X compile docs/ggov1_explainer_report.tex --outdir docs
```

The expected output is `docs/ggov1_explainer_report.pdf`.

A Signal glossary

ω	Rotor speed in per unit; 1.0 pu corresponds to 60 Hz in these simulations.
P_e	Actual electrical load power after the model’s frequency-sensitive load relation. This may differ slightly from the raw demand command.
P_m	Mechanical turbine power delivered to the rotor model.
FSRN	Candidate fuel-stroke request from the normal speed/power governor.
FSRA	Candidate request from the acceleration limiter.
FSRT	Candidate request from the temperature/load limiter.
FSR	Lowest of the three candidate requests; command sent toward the valve actuator.
v	Actual valve stroke after actuator lag, rate limits, and stroke limits.
W_f	GGOV1 fuel-flow signal. With <code>flag=1</code> , $W_f = v\omega$.

R	Droop coefficient. It determines the speed/power tradeoff in droop operation.
H	Generator inertia constant in seconds; larger H means slower frequency change for a given imbalance.
L_{dref}	Turbine load/thermal reference in per unit on turbine rating.
T_{act}	Valve-actuator time constant.
a_{set}	Acceleration-controller setpoint in per unit speed per second.
N	Number of identical, equally sharing turbine-generator units in the aggregated fleet equivalent.
D_{HCF}	Miner's cumulative damage fraction for high-frequency torsional cycles after removal of the slow torque trend.
D_{LCF}	Miner's cumulative damage fraction for slow torque-trend excursions in this study. It does not include hot-section thermal damage from starting and stopping a gas turbine.
N_f	Predicted cycles to failure at a specified equivalent stress amplitude in the assumed S-N curve.

B A novice's checklist for reading any governor plot

Before drawing a conclusion, ask these questions in order:

1. What disturbance occurred, and exactly when?
2. Did electrical load rise or fall?
3. Immediately afterward, is $P_m - P_e$ positive or negative?
4. Should frequency therefore rise or fall according to the swing equation?
5. Which candidate request is the lowest and therefore selected?
6. Does the physical valve lag behind the selected request?
7. Is a protective branch active, and if so, what physical quantity is it limiting?
8. Is the final frequency offset expected from droop, or evidence of an infeasible load or incorrect control mode?
9. Are the parameter values calibrated to the machine being discussed, or merely typical values?

References

- [1] IEEE Power & Energy Society Task Force on Turbine-Governor Modeling, *Dynamic Models for Turbine-Governors in Power System Studies*, PES-TR1, January 2013.
- [2] L. N. Hannett and A. H. Khan, "Combustion turbine dynamic model validation from tests," *IEEE Transactions on Power Systems*, vol. 8, no. 1, pp. 152–158, 1993.
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- [5] ASTM International, *ASTM E1049: Standard Practices for Cycle Counting in Fatigue Analysis*, West Conshohocken, PA.